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# Exploring student and faculty experiences and perceptions of generative artificial intelligence in medical education

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**Abstract**

The potential and challenges of using Generative artificial intelligence (GAI) in medical education are widely discussed, yet its use by medical students and faculty remains under-researched in the UK context. This exploratory sequential mixed-methods pilot study empirically investigates and compares student and faculty experiences and perceptions of using GAI in medical education. A questionnaire was developed based on focus groups and administered to undergraduate medical students and faculty within a UK medical school. Descriptive analysis and comparative analysis were used to quantify and compare student (n = 29) and faculty (n = 32) experiences and perceptions of GAI. Reflexive thematic analysis of open-ended question responses was undertaken to complement quantitative results. We found that GAI is being used for a variety of purposes by students and faculty in learning/teaching/assessment. However, both of them generally have relatively low self-confidence in GAI-related knowledge and skills. Faculty members tend to have stronger concerns of GAI limitations and its ethical challenges. Males are more confident in using GAI and have more positive attitudes towards GAI than females. The findings also provide preliminary evidence for the importance of open communication of GAI between students and faculty and the need for schools', universities', and national regulators' strategic support to ensure faculty and student ethical and effective use of GAI and equitable access to GAI technology between subgroups. Our findings and the pilot tools can inform future research investigating the use of GAI in medical education in other institutional context or at a larger scale.

**Keywords** Medical education, Generative artificial intelligence, Undergraduate students, Faculty

## 1 Introduction

Generative artificial intelligence (GAI), with its fast-paced development, has been playing an increasingly important role in higher education and also in the healthcare sector [1, 2]. According to a survey on 1,041 full-time undergraduate students in early 2025 in the UK, 88% of the students have been using GAI for assessment purposes – dramatically increased from 53% in 2024 [1]. In healthcare, as synthesised in a recent review [3],

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GAI is being used for a range of purposes such as assisting with diagnosis, predicting disease risks, patient communication, and documentation automation. This highlights the need for developing and equipping medical students with GAI-related knowledge and skills for their future career as doctors in an increasingly AI-enabled healthcare system [4, 5].

In medical education, GAI is considered to have great potential. For example, researchers suggest it can support students in self-directed and personalised learning and writing, create simulation scenarios for clinical education and assessments, help assess students and provide individualised feedback [6–11]. Emerging evidence has shown that the "anytime and anywhere" availability [9] of GAI increases students' self-directed learning motivation, as students can learn with it at any time and receive immediate feedback from it [12]. Moreover, due to GAI's free access (or relatively low cost), it can also reduce the financial barriers for students and medical schools to access important education resources such as clinical simulations [11, 13]. Researchers found that using ChatGPT (a well known GAI tool) in training on communication skills enhances medical students' confidence in breaking bad news to patients [14]. It is suggested that GAI can enhance the objectivity and consistency in assessment feedback and reduce human bias in assessment [15, 16]. Another study found that some particular GAI models (e.g. ChatGPT-o1) have shown great potential in supporting the development of medical students' clinical reasoning skills, although this evidence also suggests that they still lack the depth of patient-specific knowledge and cannot yet replace human experts [17].

Nevertheless, GAI's limitations (e.g. content bias and inaccuracy), the ethical challenges it brings (e.g. academic integrity, equitable access, and data privacy), and negative influence it might have on cognitive development and critical thinking, raise matters of concern [16, 18–23]. It is therefore sometimes suggested that assessment needs to transform through using the formats such as oral exams and group projects to prevent student from using GAI in, for example, essays, and assessments that emphasise the evaluation of high-level competencies [6, 24, 25]. Recognising both the challenges and potential opportunities that GAI brings to learning, teaching, and assessment in medicine, some researchers propose the hybrid intelligence model, which integrates AI as a tool complementary to a human expert [16, 26]. Importantly, integrating GAI in medical education and leveraging its potential effectively, ethically, and sustainably urges strategic training, policy, and guidance for students and also faculty which, however, remains widely [27–29].

As to how GAI is currently being used in medical education, extant empirical research is still rare and focuses largely on students only [30]. Emerging exploratory studies show that a significant proportion of medical students have been using GAI in learning. For example, in the study based on data collected from 10 universities across seven countries, 41% of the medical students are familiar with using large language models (a specific type of GAI) in medical education, and use it for understanding medical concepts, seeking real-time feedback, and preparing for exams [31]. Similarly, a local survey delivered to year 2 medical students in a Japanese university found that about 40% of the students had used GAI [32]. Nevertheless, there still seems to be hesitancy amongst medical students in adopting GAI [31]. It is argued that the relatively low usage may be related to individual and social factors, including students' individual concerns about GAI limitations and associated ethical challenges, and their educators' potential resistance to

this technology [33, 34]. Some studies also reveal gender differences, showing that male medical students use GAI more often than female students and have higher level of GAI-related knowledge [35, 36].

Despite this emerging evidence of medical students' use of GAI, how students use and perceive GAI and its impact in medical education is still under-researched. Moreover, there is little evidence regarding GAI use by teaching faculty in medicine and whether there are important differences between faculty and student perceptions about GAI. Hence, this pilot study reports the perspectives of both undergraduate medical students and faculty in a single UK medical school. It also investigates how these perspectives vary by gender.

### 1.1 Theoretical foundation and research questions

The technology acceptance model (TAM) suggests that users' perceptions of usefulness of the technology and the perceived ease of use, which are influenced by external variables (e.g. user characteristics, training, user support), are the two key factors that can account for users' self-efficacy and use behavior [37]. Developed based on TAM and some other models and theories (e.g. social cognitive theory [38]), the Unified Theory of Acceptance and Use of Technology (UTAUT) [39] proposes two similar factors – performance expectancy and effort expectancy, in addition to social influence (e.g. peer influence) and facilitating conditions, that have direct or indirect effects on use behavior of new technologies. As a development of TAM, UTAUT considers user characteristics such as gender, age, and experience as moderators in the relationships between users' perceptions of the technology and their usage, rather than merely determinants of perceptions. Both TAM and UTAUT and their adapted/extended versions have been employed in a wide range of research contexts, and there have been emerging studies applying these theories in investigating GAI in higher education [40, 41]. These theories are used to underpin our research exploring students' and faculty's perceptions and experiences of GAI and investigating group differences.

For a local pilot study, which has a relatively small scale of exploration, we specifically focus on some of the key elements of TAM and UTAUT. Using mixed methods, this study aims to explore and compare the experiences and perceptions of GAI in medical education for different subgroups (students vs. faculty, gender) through addressing the following research questions.

1. How have students and faculty been using GAI in medical education context?
2. How do students and faculty perceive GAI capabilities, limitations, and ethical challenges in medical education?
3. What support do students and faculty need to facilitate students' better use of GAI?

The findings of the study are intended to provide a descriptive picture that can help shape the use of GAI in medical education locally and more widely.

## 2 Methods

### 2.1 Design

As GAI in medical education is an emerging research area where existing empirical evidence and measures are limited, we employed a mixed-methods exploratory sequential design [42]. Focus groups were first undertaken to identify key themes and categories

about students' and faculty members' uses of GAI in learning/teaching/assessment and their perceptions to inform the development of a questionnaire.

This pilot study focuses on a large established medical school at a university in the UK, and potential participants include all students enrolled on the medical undergraduate program (MBChB) and all faculty members involved in the teaching/assessment at the School. This program lasts 5 years, with years 1–2 focusing on foundational and early clinical learning (i.e. preclinical stage) and years 3–5 focusing on immersed clinical learning (i.e. clinical stage). Ethical approval was obtained from the Faculty's Research Ethics Committee (ethics approval number: MREC1986) before data collection started. With gatekeeper approval, research invites were sent to all MBChB students via emails, and teaching faculty were invited via emails sent to those on the staff mailing list and posts on the Teams channel of the School of Medicine. Participation was entirely voluntary.

### 2.2 Data collection and analysis

#### *2.2.1 Focus groups*

We first conducted three focus groups (FG) – one each with preclinical-level students, clinical-level students, and faculty members during the week 18th November – 22nd November in 2024, to explore their experiences of using GAI, perceptions about GAI, and suggestions (FG guide available on request). As mentioned before, perceptions of usefulness/performance expectancy and the perceived ease of use/effort expectancy is highlighted by TAM and UTAUT. Informed by these theories and considering GAI's generally easy-to-use interfaces and extant discussions of GAI potentials and the challenges for effective and ethical use, FGs investigated participants' perceptions of GAI capabilities, perceptions of GAI limitations and ethical challenges, and also additionally explored their perceptions of the impact of GAI in medical education. Each FG discussion took about one hour at a meeting room at the School of Medicine and was audio-recorded with an encrypted University recorder with participants' consent. The recordings were transcribed by a University-approved service provider and checked and corrected (where appropriate) by the first author. There were seven preclinical-level students, three clinical-level students, and two faculty members participating in FGs.

Focus groups data were pseudonymised and then analyzed in NVivo 14 by using the reflexive thematic analysis approach (RTA) [43], which allowed us to reflexively and iteratively engage with the data to identify meaningful patterns. Following the phases suggested by this approach, we first read and re-read each transcript to familiarise with the data, followed by identifying initial codes and categorising codes into subthemes and themes. We then reviewed each theme, its subthemes, codes and coded excerpts to check the appropriateness of coding and code grouping and refine the themes, before finalising them. The themes, subthemes, and codes identified from FGs (as displayed in Appendix 1) were then used to develop the questionnaire.

#### *2.2.2 Questionnaire*

The online questionnaire was developed based on the FG data, with FG themes, subthemes, and codes translated into questionnaire sections, scales/questions, and items respectively (see the Appendices). It covered topics such as demographics, GAI use experiences, GAI use frequency (scale 0 Never-4 Daily), perceptions of GAI capabilities,

perceptions of GAI limitations, challenges, and ethical concerns (scale 1 Strongly Disagree-5 Strongly Agree), GAI impact (scale 1 Very Negative-5 Very Positive, and Don't Know), suggestions on supportive resources and activities, and self-confidence in GAI-related knowledge and skills (scale 0 Not Confident at All-4 Extremely Confident). The question on self-confidence was included, informed by TAM which suggests self-efficacy has an important role in technology use behavior. As there was no established scale on self-efficacy in doing specific GAI-related tasks, we asked about the general self-confidence instead. The questionnaire also included a small number of additional items to reflect recent discussions in literature which did not emerge from FGs. For example, we added "It lacks emotional intelligence and empathy." [44] under the question on GAI limitations and ethical challenges. In total, the questionnaire includes 19 questions (multiple-choice/Likert scale and open-ended). Open-ended questions allowed us to collect respondents' additional comments/alternative responses.

As all MBChB students and related faculty members in the focused institution were our target populations, this constrains us to do a formal piloting of the questionnaire in this research context. However, it was informally piloted through being reviewed by one medical student (FG participant) and two faculty members (one of them is FG participant), to ensure it is clear and accessible, and has captured the key content in the FG discussion. We delivered the questionnaire to wider medical students and faculty within the School and collected the data anonymously via the University's Microsoft Forms platform between January and February 2025. We received valid responses from 29 undergraduate medical students (the response rate is 2%) and 32 faculty members (the response rate is about 11%[1]), as Table 1 summarises.

Quantitative data collected via the questionnaire were analyzed in Stata 18. Reliability was first examined for the scales. The Likert scales on perceptions of GAI capabilities, perceptions of GAI limitations, challenges, and ethical concerns, and perceptions of GAI impact have good overall reliability, with Cronbach's Alpha value of 0.89, 0.82, 0.90 respectively. Due to the relatively small sample size, advanced methods (e.g. confirmatory factor analysis) were not used to further validate these scales. There were several multiple-choice questions including an "Other" option. Respondents' explanation for the "Other" options they chose was coded where appropriate and quantified in the analysis. We conducted descriptive analysis to understand the overall picture of GAI uses in medical education and the perceptions of students as well as faculty. We then conducted comparative analysis, to explore whether there are differences between students and faculty members in terms of their perceptions about GAI, and whether there

**Table 1** The questionnaire sample

| | Gender *N* (%) | | | | Age | | | | Study level *N* (%) | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Male | Female | Non-Binary | Total | Min | Max | Mean | SD | Preclinical | Clinical |
| **Students** | 12 (41.4%) | 15 (51.7%) | 2 (6.9%) | 29 (100%) | 19 | 32 | 22.24 | 2.67 | 12 (41.4%) | 17 (58.6%) |
| **Faculty members** | 11 (34.4%) | 21 (65.6%) | | 32 (100%) | 26 | 72 | 48.52 | 10.74 | | |
| **TOTAL** | 23 (37.7%) | 36 (59.0%) | 2 (3.3%) | 61 (100%) | | | | | | |

[1] The exact number of relevant faculty members is not available. This response rate was calculated based on the estimated counts of contracted staff at the School, sessional staff, and seconded staff who contribute to delivering the MBChB program in this institution.

are gender differences in responses. Independent-sample T test was conducted for the variables (average perceptions of capabilities, average perceptions of concerns, average perceptions of impact) meeting the normality assumption, while Mann-Whitney U test was employed for those that violated this assumption. Fisher's exact tests were used to compare the percentages of students and faculty in suggesting the inhibitors of starting to use GAI and in suggesting the need of specific supportive resources/activities. All responses to open-ended questions in the questionnaire were analyzed with RTA following the steps as we did for FGs to complement the quantitative results, with codes S01/S02/…/S29 and F01/F02/…/F32 indicating student respondents and faculty respondents respectively. Due to limited space, this paper will only report on and discuss the questionnaire results.

## 3 Results

We begin by outlining overall experiences of using GAI and then probe a range of perceptions about its usage in teaching and learning, and quantify gender differences in usage and perceptions, and finally present the findings on the supportive resources and activities that are needed. Particular quotes and subthemes around experiences/perceptions are also presented to illuminate the findings.

### 3.1 Student and faculty experiences of using GAI

Across all respondents, on average, their frequency of using GAI is relatively low (1.08 on a scale 0–4, i.e. *rarely*). 58.6% of medical students and 40.7% of faculty members reported that they have been using GAI to various extents in a medical education context. However, Mann-Whitney U test results do not provide sufficient evidence that the difference in the distributions of GAI use frequency between these two groups is statistically significant, although the difference has a small-to-moderate effect size ($p = 0.051$, $r = 0.25$).

Respondents' overall confidence in GAI knowledge and skills is also low (1.38 on a scale 0–4, i.e. between *slightly confident* and *moderately confident*). The distribution of students' overall confidence in their GAI knowledge and skills is not statistically significantly different from that of faculty members ($p = 0.18$, $r = 0.17$).

GAI has been used for a variety of purposes for learning, teaching, and assessment in medical education context. Students mostly use GAI to summarise specific topics (76.5%), generating questions or stations for preparing for exams (41.2%), and simulating medical cases (41.2%) (see Fig. 1). For faculty, GAI is being used more often for improving writing (53.8%) and preparing teaching materials (38.5%). It is also used by faculty to generate questions or stations for exams but is not reported to have been used in writing feedback on students' assignments or marking students' assignments.

Amongst those who have not yet used GAI, 58.3% of the students suggested that they do not trust GAI, which is significantly higher than that of faculty members (15.8%) ($p < 0.05$). 47.4% of faculty members reported that they have had no related training, and 42.1% suggested that they have had no time to explore GAI. Being unsure about the regulations around the use of GAI is another factor inhibiting both students (33.3%) and faculty (31.6%) from using it. 31.6% of faculty members additionally commented that GAI was not relevant to their work.

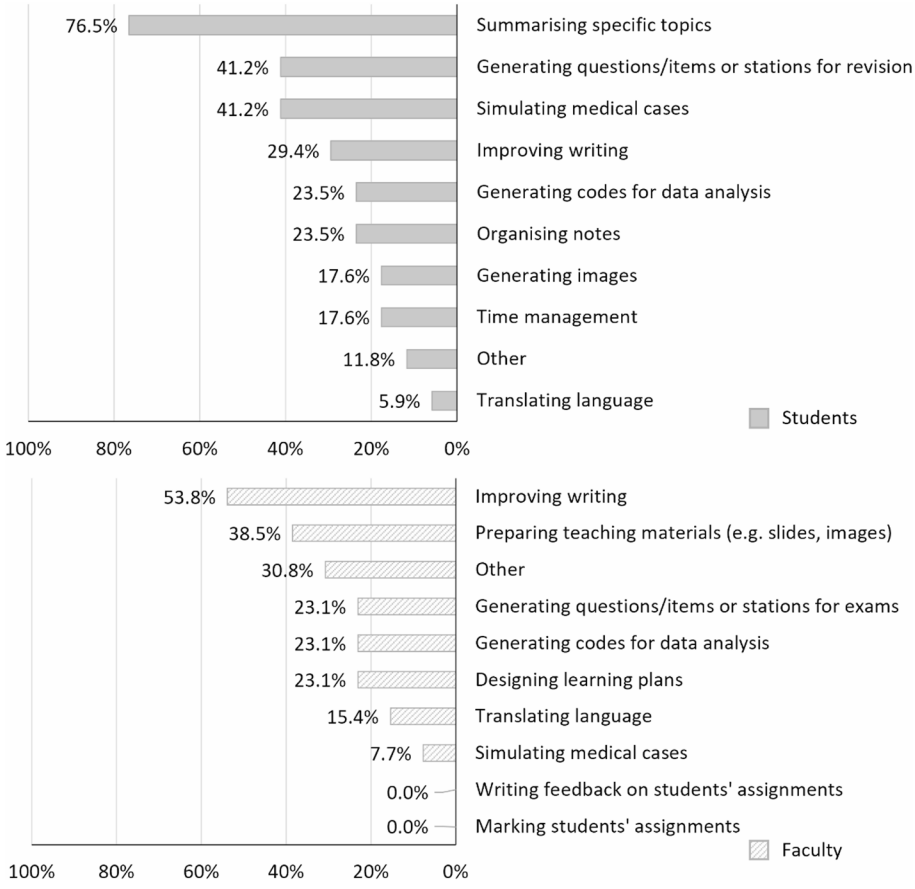



**Fig. 1** Uses of GAI

### 3.2 Student and faculty perceptions of GAI

#### *3.2.1 GAI capabilities*

Amongst all students and faculty members, on average, they slightly agreed (mean = 3.43 on the scale 1–5) that GAI has capabilities in medical education, and particularly agreed that ***it saves time*** (mean = 3.92) (see Table 2, ordered by most to least positive average response overall). As students commented, GAI "makes life easier" (student respondent S21) and "can be used to save time in regard to making summaries" (S06). There is no statistically significant difference between students' perceptions of GAI overall capabilities and those of faculty members ($p = 0.16$, Cohen's d = 0.37). Compared with faculty, students had more positive attitudes towards GAI's capabilities in making some medical education resources more[2] financially accessible ($p < 0.05$, $r = 0.26$)[2].

Respondents additionally commented on the ***effectiveness of using GAI capabilities***. Two of them suggested that GAI can facilitate effective learning or teaching. For example, one faculty member said:

> *It has an enormous potential to help and inform both students on the course and staff to be more effective, if implemented properly and with an open mind. (F16)*

[2] Independent-samples T tests were conducted for comparing student and faculty average perceptions of GAI capabilities, GAI limitations, challenges, and ethical concerns, and also GAI impact. Mann-Whitney U tests were conducted for comparing perceptions of specific capabilities/limitations, challenges, and ethical concerns/impact for the two groups.

**Table 2** Perceptions of GAI

| Perceptions of GAI | Overall mean | Mean Difference (Students – Faculty) | Significance test Two-Sided *p* | Effect size |
|---|---|---|---|---|
| *GAI capabilities* | | | | |
| • It saves time. | 3.92 | 0.36 | 0.23 | 0.16 |
| • It makes medical concepts more understandable. | 3.63 | 0.55 | 0.07 | 0.23 |
| • It makes some otherwise expensive medical education resources more accessible. | 3.46 | 0.58 | < 0.05 | 0.26 |
| • It helps reduce language barriers to medical education. | 3.45 | −0.10 | 0.71 | 0.05 |
| • It increases learning productivity. | 3.31 | 0.44 | 0.17 | 0.18 |
| • It motivates learning. | 2.95 | 0.10 | 0.78 | 0.04 |
| Capabilities_mean | 3.43 | 0.34 | 0.16 | 0.37 |
| *GAI limitations, challenges, and ethical concerns* | | | | |
| • It can generate inaccurate content. | 4.44 | −0.32 | 0.11 | 0.21 |
| • It lacks transparency on its sources of training data and algorithms. | 4.17 | −0.60 | < 0.05 | 0.26 |
| • It can generate biased content. | 4.16 | −0.44 | 0.05 | 0.25 |
| • It causes copyright concerns. | 4.15 | −0.35 | 0.17 | 0.18 |
| • It brings threats to academic integrity. | 4.12 | −0.16 | 0.75 | 0.04 |
| • It lacks emotional intelligence and empathy. | 3.85 | 0.35 | 0.45 | 0.10 |
| • Its development lacks consideration of environmental sustainability. | 3.81 | −0.21 | 0.52 | 0.09 |
| • It increases data security risks. | 3.80 | −0.09 | 0.65 | 0.06 |
| • The access to GAI tools is inequitable in the world. | 3.79 | −0.24 | 0.27 | 0.15 |
| • It encourages superficial learning. | 3.67 | −0.56 | 0.13 | 0.19 |
| • The access to GAI technology is inequitable amongst students at the School. | 3.21 | −0.68 | < 0.05 | −0.28 |
| Concerns_mean | 3.93 | −0.33 | < 0.05 | −0.55 |

| | | | | | *n* (%) of the response "Don't Know" | |
|---|---|---|---|---|---|---|
| | | | | | Students | Faculty |
| *GAI impact* | | | | | | |
| • Curriculum content | 3.50 | −0.22 | 0.41 | 0.15 | 11 (37.9%) | 14 (43.8%) |
| • Teaching methods | 3.26 | 0.32 | 0.44 | 0.12 | 11 (37.9%) | 8 (25.0%) |
| • Learning methods | 3.24 | 0.79 | < 0.05 | 0.30 | 8 (27.6%) | 8 (25.0%) |
| • Career preferences | 3.06 | −0.14 | 0.48 | 0.15 | 8 (27.6%) | 17 (53.1%) |

**Table 2** (continued)

| | | | | | *n* (%) of the response "Don't Know" | |
|---|---|---|---|---|---|---|
| | | | | | **Students** | **Faculty** |
| • Equal access | 2.81 | 0.38 | 0.23 | 0.19 | 6 (20.7%) | 10 (31.3%) |
| • Assessment programme design | 2.74 | −0.21 | 0.7 | 0.07 | 13 (44.8%) | 13 (40.6%) |
| • Professionalism development | 2.73 | 0.01 | 0.95 | 0.01 | 7 (24.1%) | 9 (28.1%) |
| • Assessment methods (i.e. how we assess) | 2.53 | 0.26 | 0.38 | 0.15 | 9 (31%) | 14 (43.8%) |
| • Assessment content (i.e. what we assess) | 2.67 | 0.33 | 0.17 | 0.25 | 10 (34.5%) | 14 (43.8%) |
| • Fairness of summative assessment | 2.48 | 0.24 | 0.2 | 0.20 | 6 (20.7%) | 11 (34.4%) |
| • Academic integrity in summative assessment | 2.29 | 0.41 | 0.07 | 0.27 | 6 (20.7%) | 8 (25.0%) |
| • Authenticity of assessment | 2.18 | −0.31 | 0.45 | 0.13 | 8 (27.6%) | 14 (43.8%) |
| Impact_mean | 2.77 | 0.17 | 0.44 | 0.21 | | |

*Student S24 also pointed out that "it should be used as a helpful resource not a replacement for our own work". One of the faculty respondents who has not yet used GAI commented that they were "yet to see convincing evidence that GAI improves quality of teaching and learning" and suggested it "has to be used responsibly and sensibly" (F22).*

More respondents argued that ***effectiveness depends on use.***

*I think the effectiveness of GAI is what you make of it, it's very easy to use it poorly, … and fairly difficult to master. (S08)*[3]

*The question is less about the capabilities of GAI, and much more about the design of the processes it's involved in or the care taken in the application of its outputs. A badly designed learning activity is going to remain badly designed whatever technologies are brought to bear on it. (F01)*

*To achieve effective and appropriate use of GAI, F11 suggested that people need to "understand its capabilities and potential pitfalls", whilst "educators & students have started using it without necessarily understanding it or having guidelines and boundaries for appropriate use in HE".*

#### *3.2.2 GAI limitations, challenges, and ethical concerns*

On average, respondents generally agreed that there are various limitations, challenges, and ethical concerns around GAI in medical education (mean = 3.93 on the scale 1–5) - Table 2. Inaccuracy, lack of transparency, and bias are the top three concerns. Faculty members have statistically significantly stronger perceptions about GAI limitations and ethical challenges than students ($p < 0.05$, Cohen's d = −0.55). Compared with students, faculty members particularly had higher concerns about students' inequitable access to GAI technology and the lack of transparency on GAI sources of training data and algorithms ($p < 0.05$, $r = 0.28$; $p < 0.05$, $r = 0.26$). They also tend to have higher concerns about bias although the statistical significance of the difference is at the borderline level ($p = 0.05$, $r = 0.25$).

Respondents, especially faculty members and those who have never used GAI in medical education, particularly suggested that GAI can bring ***potential dangers to learning***. To some extent, because it saves time and is "the 'easy option'" (F21), it "could become a crutch of sorts" (S11), which "can easily create more problems than it solves" and "destroy creativity and innovation if it becomes more widespread" (F22). Respondents "worry that it takes away a student's interest in doing their own study/research" (F21) and is "quite dangerous to make people lazy rather than properly learning concepts" (F29). It may also "encourage students and staff not to develop their problem-solving skills and become more insightful practitioners" (S20). Student respondent S11 also said:

*I believe GAI has passed the point of being a tool, and is instead more of a machine that carries out tasks for us, instead of with us, and I believe this has the potential to damage a lot of what makes human learning and even culture special. (S11)*

[3] Independent-samples T tests were conducted for comparing male and female average perceptions of GAI capabilities, GAI limitations, challenges, and ethical concerns, and also GAI impact. Mann-Whitney U tests were conducted for comparing GAI use frequency and overall confidence for the two groups.

With this concern, a small number of respondents were totally against using of GAI for learning, commenting as below.

> *I will be extremely disappointed if the university decided to use this software in the future. It is harmful to learning culture, and the global environment. It is deeply unethical and stands completely in the face of what doctors should believe in. (S29)*
> *It is a total barrier to understanding … telling them that while it might be useful for them ultimately once they have completed their education, not to use it at all as part of their learning. (F27)*

F17, although having not used GAI, had a different view and suggested that students should embrace GAI.

> *I think students should embrace it, analyse it and troubleshoot it. Treat it like a future patient that needs diagnosis, and long-term management to enhance it's well-being. There are synergies between humans and GAI and those should be embraced. (F17)*

Some respondents also considered that GAI ***lacks depth and expertise*** in generating content. Faculty member F23 pointed out that "it doesn't have access to scientific literature", which "is a huge limitation" and "a barrier that makes results superficial". Student S06 suggested:

> *it doesn't have the ability to understand complicated concepts that aren't widely discussed in its database (online) and has no ability to do logical reasoning of concepts or critical analysis of research. (S06)*

Unlike "actual doctors who take their time to teach us and provide case based experience" (S20), it is considered that GAI "will only provide one view point and miss many nuanced aspects that are so extremely important in medical education" (S29) and be "likely to undermine the expertise and intellectual property needed to develop engaging, relevant and innovative learning" (F26).

The lack of genuine emotion in GAI was commented on by some respondents. It is considered that GAI's "empathy is a bit robotic" (S27) as empathy or emotional intelligence "seem to be products of a level of processing that AI can only mimic" (F09). Hence, one respondent thought "using AI exclusively for medical education will not be effective in teaching empathy and humane interventions when practicing" (S12).

Nevertheless, it is suggested that newer versions of GAI have had some development in this aspect. As a student added:

> *… but later generations of ChatGPT, for example, you can talk to and it will show some insight into sensitive topics. (S27)*

Participant F14 commented, perhaps with a degree of sarcasm, that "even the free versions are more emotionally intelligent and empathetic than many of my peers".

F09, however, suggested that lack of empathy or emotional intelligence is not an issue exclusive to GAI, as "many educational resources are largely devoid of empathy". The critical role of users was highlighted by F12 who argued that "AI itself is not emotional but the user is".

#### *3.2.3 GAI impact*

In terms of perceptions of GAI impact, a number of the respondents responded that they were ***not clear*** whether GAI is impacting on different aspects of medical education positively or negatively (see Table 2). As F01 commented, "I don't think the picture is anywhere near clear enough to indicate what's happening in any of these areas with absolute clarity". It is also considered that its impact is "not quantifiable", since all the areas presented in Table 2 "are affected by a balance between the benefits that GAI can offer to enhancing education vs. the downsides of students and staff cutting corners/using inaccurate and/or poor quality resources" (F10).

On average, students' and faculty's overall perceptions of impact are not statistically significantly different ($p > 0.05$, Cohen's d = 0.21). They generally perceived that GAI has a slightly negative impact on most of aspects of medical education, except for curriculum content, teaching methods, career preferences (i.e. mean < = 3 on the scale 1–5).

In terms of its impact on learning methods, students' attitude is more positive than faculty members and the effect size is moderate ($p < 0.05$, $r = 0.30$). In terms of its impact on assessment, faculty members tend to be more worried about GAI influencing academic integrity in summative assessment negatively; however, such attitudes of the two groups have no statistically significant difference ($p = 0.07$, $r = 0.27$). Due to the relatively small sample size of the valid responses on this scale, these results however need to be interpreted with caution.

Respondents additionally commented on GAI's ***impact on assessment***. F08 expressed their concern about the potential negative impact on the validity of assessment for medical students as "someone may use GAI to write assessments and go undetected" while "knowledge is a key requisite for being a doctor". S11 added that "every single assessment which doesn't have intense invigilation or blocks tabbing is at risk/already is compromised by GAI".

Other respondents suggested rethinking and reshaping of assessment approaches to "make them more authentic" (F12). It was also suggested that there is a need to consider "whether written work is now possible as part of this" and "how to promote and assess critical thinking in a way that cannot be circumvented by GAI" (F25). Another faculty member F05, however, thought "academics are unconfident and not equipped to develop the skills to develop authentic assessment that uses AI explicitly". F01 added somewhat obliquely:

> *I think there are definitely problems that will be posed in regard to assessment, but this relates in part to the lazy approaches to assessment that have been endemic in Universities (and Medicine in particular) for many years. Some of those problems will, I think, be overdue spurs for innovation and rethinking how these aspects of the curriculum work. (F01)*

They further added that University's response will determine whether the impact is "positive" or "negative" (F01).

### 3.3 Gender differences

Amongst all students and faculty members, males tend to use GAI more often ($p < 0.01$, $r = 0.37$), have higher overall confidence in their GAI knowledge and skills ($p < 0.001$, $r = 0.50$), fewer concerns about GAI limitations and challenges ($p < 0.01$, Cohen's d

= −0.85), and have more positive attitudes towards GAI's impact on medical education ($p < 0.05$, Cohen's d = 0.59). There is no statistically significant gender difference in respondents' perceptions of GAI capabilities ($p > 0.05$, Cohen's d = 0.29) (Table 3).

### 3.4 The need of supportive resources and activities

Students reported that in order to support effective, responsible, ethical, and equitable use of GAI open communication between them and educators about use of GAI is needed (62.1%), followed by clearer regulations and guidance from the Institution (58.6%) and student training sessions on how to use GAI (55.2%) (Table 4). The majority of faculty (90.6%) called for training for them on how to use GAI. Most faculty members also suggested the need of open communication with students, clearer regulations and guidance and policy evaluation and update, training for students (78.1%–87.5%). 62.5% of them also suggested the need of advice and expertise support from national advisory bodies. Based on Fisher's exact tests, faculty members expressed greater need for these types of resources and activities more than did students (see Table 4).

In related open comments, some faculty respondents also suggested the need for GAI-related training specifically for "HE and medicine/healthcare professional education" (F05) and "on assessment design and marking" (F12). It is considered that "knowing how to use GAI safely and appropriately makes it a great tool" (F02) and it "must be applied to all who may potentially use it" (F28). For students, F05 stated that currently "We are allowing students to use GAI without supporting them to develop the skills needed to use it effectively and responsibly" and suggested the need to incorporate AI training in the existing curriculum.

> *I feel like the focus is on efficiency saving with no real consideration of the need to develop critical thinking skills about the content generated. I would like to see a strand of teaching on the role of tech in medicine and to have a discussion about AI as part of that (benefits, risks, ethics, implications for health inequalities etc.). (F05)*

F09 suggested the need for some more examples – "discussions of GAI are introduced in conjunction with an opportunity to apply those principles/ideas, such as an assignment or workshop".

For some faculty, it is important to have more support for developing faculty understanding about AI. As F05 said:

**Table 3** Gender difference in experiences and perceptions of GAI

| | Gender | *n* | Mean | SD | SE | Mean difference | Significance test Two-sided *p* | Effect size |
|---|---|---|---|---|---|---|---|---|
| Use Frequency | Male | 23 | 1.70 | 1.46 | 0.30 | 1.03 | < 0.01 | 0.37 |
| | Female | 36 | 0.67 | 1.07 | 0.18 | | | |
| Overall Confidence | Male | 23 | 2.13 | 1.25 | 0.26 | 1.33 | < 0.001 | 0.50 |
| | Female | 36 | 0.81 | 0.98 | 0.16 | | | |
| Capabilities | Male | 23 | 3.58 | 0.90 | 0.19 | 0.27 | 0.28 | 0.29 |
| | Female | 35 | 3.31 | 0.95 | 0.16 | | | |
| Challenges/ limitations/ethical concerns | Male | 23 | 3.66 | 0.52 | 0.11 | −0.48 | < 0.01 | −0.85 |
| | Female | 36 | 4.13 | 0.59 | 0.10 | | | |
| Impact | Male | 21 | 3.06 | 0.73 | 0.16 | 0.49 | < 0.05 | 0.59 |
| | Female | 33 | 2.57 | 0.87 | 0.15 | | | |

**Table 4** Supportive resources and activities needed

| | Student training on how to use GAI | Faculty training on how to use GAI | Student training on GAI-related ethics | Faculty training on GAI-related ethics | Institution's clearer regulation and guidance | National advisory body providing advice and expertise support | Student-educator open communication | Continuous evaluation and updates of related policies | Other |
|---|---|---|---|---|---|---|---|---|---|
| Students | 55.2% | 37.9% | 44.8% | 41.4% | 58.6% | 31.0% | 62.1% | 44.8% | 6.9% |
| Faculty | 87.5% | 90.6% | 78.1% | 78.1% | 84.4% | 62.5% | 84.4% | 81.3% | 15.6% |
| Sig of difference (Two-Sided p) | <0.01 | <0.01 | <0.01 | <0.01 | <0.05 | <0.05 | 0.08 | <0.01 | 0.43 |

> *Often with any innovation in med ed staff opt in, in the case of AI and the implications for academic integrity I think there needs to be more effort to help staff to understand it is here and they need to understand it. Many colleagues seem to be in denial that students are not using it and basing assessment design on a context that no longer exists. (F05)*

Another respondent (F16) perceived stigma by faculty of GAI student users, which hinders students' appropriate use of GAI. Therefore, they suggested that faculty also need to enhance their understanding about GAI capabilities for student learning.

> *If students use it in an environment were using it is not welcomed, then they are likely to misuse it and can end up plagiarising etc. (F16)*
> *Many staff assume the only way GAI can be used by students is to generate an entire answer or essay, whereas this is far from the truth. Everybody should be aware of the capabilities of GAI and understand how it can benefits students on their university journey. (F16)*

Speaking to F16's comment about the educational environment, others call for a cultural shift regarding GAI use. As F04 commented:

> *There needs to be a cultural shift regarding its use. Current distrust across some of the academic team needs to be challenged through discussion and education. The main limitations we have are the differences in speed that individuals are embracing it and this not fitting with students' timeframes.*

This, as suggested by F01, needs clear direction and vision and vigour from the University and its Faculties/Schools.

> *GAI is one of the biggest challenges to the way that universities operate in a long time. It's possibility to be a disruptive injection of change for the good is just as high as its possibility to be destructive, and it will be the leadership and vision of the University and its Faculties/Schools that will determine exactly which way it goes. (F01)*

## 4 Discussion

This study investigated the GAI use experiences of both undergraduate medical students and faculty in a medical school in the UK and their perceptions of GAI in medical education. Our findings provide empirical evidence revealing the variety of GAI uses in learning, teaching, and assessment in medical education, and also suggesting some differences between students and faculty in their GAI use experiences and perceptions. It contributes to extant discussions on GAI's potential benefits in supporting students' learning and increasing faculty's work efficiency [10]. It also provides insights into developing targeted training sessions for students and faculty for specific use purposes, and designing innovative teaching sessions/assignments which integrate GAI to develop student skills such as navigating and synthesis, evaluation and appraisal, and clinical reasoning.

### 4.1 Key findings

We found more than half of the students have been using GAI in their learning. This is higher than that found amongst medical students in the contexts in published studies

[31, 32]. However, it is much lower than that reported by the UK national-wide survey on undergraduate students across different disciplines, which suggests that almost all students have been using GAI, with a majority of them having used it for assessments specifically [1]. This suggests there might be differences between disciplines in GAI adoption. Some medical students in our study have hesitancy in adopting GAI [32], which appears due to them not trusting GAI enough or having insufficient social resources (e.g. training, clear regulations) to support their GAI use [33, 34].

Adding to extant literature that focusses on students, our study found resistance to the use of GAI in medical education amongst some faculty members. This resistance, which might be associated with their perceptions of GAI, may limit opportunities to explore how GAI could support learning and teaching practices in medical education. As reported by faculty, lack of training, time, clear regulation, and relevance are the main factors inhibiting their use of GAI. A number of studies include a general call for GAI-related training for students and sometimes for faculty [28, 29] - and our data suggest that, certainly in the case of faculty, additional training is needed in this area. At the time of our data collection, there had been a degree of institution-level staff guidance and some introductory training on GAI use. Nevertheless, our findings underscore the additional need to develop faculty's AI literacy and GAI-related self-confidence, increase awareness of the growing important role of GAI in medical education, and ensure that faculty has protected time to engage in related training.

In terms of perceptions about GAI, our questionnaire respondents, especially students, generally have positive attitudes towards GAI's capabilities in making medical concepts more understandable and making some medical education resources more financially accessible, which aligns with much of the literature [11, 13, 31]. However, our student respondents generally have a neutral rather than positive attitude towards GAI's capability in increasing learning motivation [12]. In general, both students and faculty agree that GAI can save time, suggesting GAI's capability for increasing learning/work efficiency is recognised in our respondents [9]. Our findings also highlight that GAI effectiveness depends on how the users apply GAI and its outputs in their learning/work, which also further underlines the importance of training for students and also faculty for enabling them effectively to do so.

Both students and faculty members also have a variety of concerns about the limitations of GAI and ethical challenges it brings - which is generally consistent with the literature [16, 22]. Faculty tend to have stronger concerns compared to students, which helps explain the resistance that some of them have towards GAI. Respondents' concerns about GAI's potential "danger" to learning indicate caution in some respondents in the over-reliance on GAI and highlight the need for its appropriate use. That these worries are mainly from faculty members and those who have not yet had experience of using GAI in medical education context might indicate that there are elements of stereotyping of GAI and/or GAI users from non-users. Resonating with previous studies [17], some respondents in our study also perceived that GAI lacks sufficient depth and expertise. However, we also witness voices suggesting the need to embrace the synergies between humans and GAI and consider that GAI "needs diagnosis, and long-term management" to enhance its performance. This highlights the importance of the nature of the interaction between human and GAI, and the critical skills needed to maximise the effective use of GAI. It also underlines the active role users must take in using and shaping GAI

purposefully, which is subtly different from the "hybrid intelligence" model of usage as suggested by some researchers [26].

Our data suggest that students and faculty see the potential of GAI in bringing innovation to curriculum content and teaching methods in medical education. Faculty's less positive attitude towards GAI's impact on learning methods as compared with students resonates with their concerns of GAI's potential danger to positive learning practices. However, a number of our respondents also consider the impact of GAI in medical education is still emergent, and this relates to their broader perceptions of current GAI capabilities and limitations. This is in concordance with other studies suggesting students' uncertainty about the impact of GAI in medical education [14, 34].

Some of our respondents had concerns about the potentially negative impact of GAI on assessment integrity and also with regards to assessment authenticity. Their suggestions for transforming assessment in this new context are broadly consistent with literature [6, 24, 25]. However, this might indicate the need to develop faculty capabilities and knowledge in this area - our findings highlight that it requires institution-level initiatives to maximise the positive impact of the use of GAI in assessment and in medical education more generally.

In terms of gender, we found some evidence of males using GAI more frequently than females, with higher confidence and more positive attitude about its usage in teaching and learning – confirming previous work in this area [35, 36]. These findings may indicate differences in confidence or usage patterns across genders, which should be explored further in larger studies to provide insights into the equality of GAI access across subgroups as the technology becomes more embedded within medical education practices.

Finally, our work adds to the extant studies which highlight the need for developing supportive resources and appropriate training materials in the use of GAI within medical education [27, 28] – for both faculty and students, and also underscores the need for open communication on GAI use between students and faculty. Ideally, these resources would allow scope for individual agency in the use and implementation of these new technologies. We would also argue that this may be best achieved through a synergy of various stakeholder views (i.e. students, faculty, patients, medical schools, universities, and national regulators) to ensure that the growing use of GAI in medical education develops in ethical and equitable ways that enhance student education experience to develop future medical practitioners.

### 4.2 Study limitations

This study is situated in a single medical school with a relatively small sample which may limit its generalisability in capturing the full range of uses of GAI across all medical students, faculty in the medical school and more widely. Moreover, although we highlighted that all MBChB students and relevant faculty members, whether they had any GAI use experiences or not, were eligible to participate in this study, there remains the possibility of some self-selection bias in the data of the participants, which might then limit the representativeness of participants. It is noted that this study has relatively small FG sizes which may limit the qualitative FG data. To address this limitation, we developed the questionnaire based on FG data whilst also taking into consideration of wider relevant topics/issues discussed in recent literature. It is also acknowledged that the scales in the

questionnaire were not further validated due to the relatively small sample size (as we have mentioned in Methods). However, the reliability was examined and found to be good which provides emergent evidence of their methodological quality. As the University updates GAI-related policies and resources (e.g. recently has made all staff and students complete mandatory online training on using GAI with more and more themed courses on GAI available), students' and faculty's experiences of GAI will evolve in the future. However, as an exploratory pilot study, it does highlight a range of perceptions around GAI and its use in medical education, differing needs for support, and indications of how these might vary across sub-groups (e.g. students/faculty and male/female), which might be relevant in other institutional/country contexts.

## 5 Conclusion

Drawing on conceptual elements from TAM and UTAUT, this study provides a descriptive picture of the usage and perceptions of GAI in medical education. Our emergent findings of differences between faculty and students, and some by gender underline the importance of generating better understanding of such a rapidly changing teaching and learning landscape, This suggests that there is indeed a need for national-scale research to capture the overall picture of GAI use in medical education. This would include the need to understand how more equitable access to and ethical use of GAI going forwards can be developed. Conceptually, such a study would also need to be based on relevant theories (e.g. UTAUT [45]) to explore the differences in uses and needs between a range of different subgroups and the mechanism driving/inhibiting GAI uptake. Such work would help inform national-level regulations, policies and support needs. Ideally, such research would go beyond just medical education to investigate similarities and differences with other HE disciplines, especially across other health professions. Given the fast-paced nature of GAI development and the evolving landscape of GAI use, it would be beneficial for the national research to enable a legacy of continuous evaluation as GAI use grows – both within and without medical education.

## Supplementary Information

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Supplementary Material 1

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**Author contributions**
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**Data availability**
Due to confidentiality and ethical considerations, the data are not publicly available. Relevant qualitative data supporting the findings are included in the article.

## Declarations

**Ethics approval and consent to participate**
Ethical approval was obtained from the University of Leeds's Faculty of Medicine and Health Research Ethics Committee (FREC) for School of Medicine (ethical approval number: MREC1986) before data collection started. The study was

conducted in accordance with the University's research ethics guidelines applicable when human participants are involved. Informed consent was obtained from all participants prior to their involvement in this study.

**Consent for publication**
Participants provided consent for publishing findings of anonymised data derived from this study.

**Competing interests**
The authors declare no competing interests.

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